INCH-POUND

MIL-C-63513A (AR)
AMENDMENT 6
19 August 1991
SUPERSEDING
AMENDMENT 5
27 July 1991

MILITARY SPECIFICATION

CARTRIDGE, 105MM, TP-T, M490A1,
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-63513A (AR), dated 5 September 1985 and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

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* 2.1.2: Add the following under PRODUCT AND PACKAGING DRAWINGS:

"12935042 - Case, Cartridge, and Liner Assembly (SWCC)
12935040 - Cartridge, 105mm, TP-T, M490A1, (SWCC)".

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* 3.2: Delete in its entirety and substitute:

"3.2 Cartridges. The cartridges shall comply with all requirements specified on Drawing (dwg.) 9343009-1 or 9343009-2 or 12935040 (alternative) and with all requirements specified in applicable specifications."

* 3.3 Between last two words, "propellant" and "used", add: "and type of cartridge case".

* 3.5.1: Delete in its entirety and substitute:

"3.5.1 Muzzle velocity. The corrected mean muzzle velocity shall be not greater than 3885 feet per second (fps), (1184.15 meters per second (mps)) or less than 3815 fps (1162.81 mps) and the standard deviation for the lot shall be not greater than 9.5 fps."

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* Add paragraph 3.9:

"3.9 Cartridge case crimp (bullet pull). Drawing 9343009-1 or 9343009-2. The cartridge case shall separate from the projectile at a load between 4000 and 6000 pounds.

Drawing 12935040 - The cartridge case shall separate from the projectile at a load between 2300 and 6000 pounds."

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* Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

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* Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

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4.4.2.1:

* Next Higher Assembly: Delete "9343010-1 or 9343010-2" and substitute "9343010-1 or 9343010-2/12935042".

Delete drawing number "9343011" and substitute "9343011 or 12934834".

Minor 202: Delete "Evidence of poor workmanship, including liner film torn or damaged" and substitute "Evidence of poor workmanship, including torn or damaged liner film, skirt or string".

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4.4.2.2:

* Drawing Number: Delete "9343010-1 or 9343010-2" and substitute "9343010-1 or 9343010-2/12935042".

* Next Higher Assembly: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Add Major 106: "106 Evidence of contamination in the inside diameter of the cartridge case 100% 3.8 Visual".

Change "Length of liner incorrect (Flush to 1/4 inch above mouth of case)" to "Length of liner incorrect) Flush to 1/2 inch above mouth of case)."

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4.4.2.3:

* Next Higher Assembly: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Critical Defect: Add: "2 Plug missing 100% 3.2 Visual".

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4.4.2.4:

* Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Delete: "201 Evidence of poor workmanship 1.65% 3.8 Visual" and substitute: "201 Evidence of poor workmanship 0.65% 3.8 Visual".

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4.4.2.5:

* Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Major 102: Delete Examination or Test column in its entirety and substitute: "Liner flaps not folded over propellant and cemented or secured with draw strings".

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Major 103: After examination or test item, add "(Note 1)".

Add Major 104: "104 Liner skirts not tightened over propellant with proper knot 100% 3.2 Visual".

Add Note 1: "1. Only one inspection is applicable depending on the method of closure utilized."

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* 4.4.2.6: Next Higher Assembly: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

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4.4.2.7:

* Drawing Number: Delete "9343009-1 or 9343009-2" and substitute "9343009-1 or 9343009-2/12935040".

Delete: "104 Gap between cartridge case and obturator, max. 0.40% 3.2 Gage" and substitute: "104 Gap between cartridge case and obturator, max. 100% 3.2 Gage".

Add Major 105: "105 Forward bourrelet diameter, max. (Note 4) 100% 3.1 Ring Gage".

Add Major 106: "106 Runout of cartridge assembly NA 3.2 Gage/4.5.5.1".

Add Note 4: "4. To be performed immediately prior to packaging."

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* 4.4.3.4.1 Delete in its entirety and substitute:

"Muzzle velocity. The lot shall be rejected if the standard deviation of the sample cartridges multiplied by the factor 0.70 exceeds the lot requirement."

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4.5.1.2: Add new paragraph 4.5.1.2 as follows:

"4.5.1.2 Contamination in the inside diameter Prior to assembly of liner to cartridge case, solvent wipe (Acetone, Spec O-A-5) inside diameter where the adhesive/liner touches the cartridge case. Solvent and wiper (lint free) shall be visually clean."

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4.5.5 Delete the entire paragraph and substitute the following:

"4.5.5 Profile and alignment check. The profile and alignment of each crimped cartridge shall be verified by a gage corresponding to drawing 8828098, which is to be horizontal ($0^{\circ} \pm 2.5^{\circ}$). The rear face of the cartridge shall be positioned within one inch of the gage's push arm and be inserted with a peak force not to exceed 60 pounds (27.2 kg) and a peak velocity of 2 ft/sec (61 cm/sec). No manual assistance (i.e., rotation, agitation, etc.) is permitted during inspection. Any cartridge which fails to be inserted or seated with the required force and velocity in any orientation and in accordance with the following sampling plan shall be rejected:

a. Inspect (50) cartridges in 4 angular orientations approximately 90° apart.

b. After (50) consecutive cartridges successfully chambergage, switch to chambergaging 1 out of 100 in 4 angular orientations and all others in 1 (one) random orientation.

c. Any failure will result in returning to step (a.).

When applicable, rejected cartridges which are reworked or repaired in accordance with Government approved procedures shall be reinspected in 4 angular orientations approximately 90° apart. Failure of cartridge to be inserted or seated in any orientation shall be cause for rejection of the cartridge. The profile and alignment gage shall not be utilized as an alignment tool."

Add the following paragraph:

"4.5.5.1 Runout inspection. Any cartridge that exceeds the runout dimension per drawing 9343009 shall be chambergaged in 4 angular orientations approximately 90 degrees apart (see 4.5.5). This feature shall be controlled using Statistical Process Controls in accordance with the Contractor's government approved plan."

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